Cancer Alley: A Legacy of Environmental Injustice and Its Impact on Future Generations

Introduction:

As everyone knows, there is a very saddening story regarding the cancer alley as it still looms over us today as one of the most atrocious environmental injustices that has been captured. In this report, we will look into the history of cancer alleys of St.James Parish, Baton Rouge, and New Orleans, the demographics of the place, the toxins released as well as what are the communities fighting back. In the end we are looking into how a land of good hope became a inhabitable disaster.

History of Cancer Alley:

Since the 1980's, there was a massive influx of polluting facilities constructed next to people of color and poor communities, and what would be seen decades to come for the communities was just utmost misery and suffering. Multitude of fields of companies from oil refiners such as Shell to chemical manufacturing companies such as Dow Chemical were casting chemicals, air pollution and waste to the side of the Mississippi River. Furthermore, the influence of these companies were overwhelming which can be seen when the ExxonMobile started pumping chemicals to the Mississippi River that echoed shockwaves across the communities health surge through air pollution generation. In addition, in 1957 the Dow Chemical evicted the families around them to create a natural buffer of impact, whereas Noranda Alumina, one of the major contributors of mercury pollutants was "gifted" a loan for the creation of a new pond to dump their hazardous waste by the Louisiana Economic Development Cooperation.

After the American civil rights, the former slaves settled down next to the places that they were enslaved that formed unincorporated communities among them. Due to Jim Crow era policies and the requirement to build a new drainage system, white people in New Orleans force African Americans to relegated zones of poorly designed drainage systems. Thereby engineering and elevation of white people zones as well as Jim Crow's law pushed African Americans to live in unhygienic areas that led them to be susceptible to high risk of malaria infections and typhoid leading to higher deaths. In the journal article written by Idna G. Castellón in "*Cancer Alley and the Fight Against Environmental Racism*" found that after Hurricane Katrina ravaged through the plains of Louisiana, only 53% of African Americans were able to move back home as compared to white people with 81% *(Castellón 2021)*. Furthemore, looking at the political situation of the parish, the majority of African Americans lived in unincorporated communities but the parish was located next to those communities. The parish had governance and policies to govern their communities and as petrochemical or plastic companies came knocking the door, the parish allowed them to construct right next to the unincorporated communities. The example could be seen in the journal by Idna G. Castellón in "*Cancer Alley and the Fight Against Environmental Racism"* that predominant whites of St.John Council in parish "rezoned" the areas next to the unincorporated communities and that allowed the petrochemical Formosa to flourish *(Castellón 2021)*.

EPA and other agencies discrimination and their incompetence:

Now you might be wondering where the government was, where were the laws, why was EPA not protecting the people ? Well, further investigation showed that the Congress did pass federal laws to protect citizens from pollution, improper enforcement of those laws by the EPA led to even more environmental racism in Cancer Alley but first, let us understand what were the laws/acts that were passed.

The Clean Air Act (CAA) was introduced in 1963 which regulates pollutants such as ground-level ozone, carbon monoxide, lead, particulate matter, sulfur dioxide and nitrogen dioxide. These regulations led to EPA establishing National Ambient Air Quality Standards (NCAA) for each pollutant and limit on atmospheric concentration of the pollutant. The EPA deploys a two step process to see the impact of hazardous pollutants generated by the industrial facilities, for one, the EPA implements a technology standard that can regular complaints source emission level, and then after eight years goes by, the EPA assesses the health risk from those source emissions from each group. When the CAA inspector general conducted an assessment of CAA law enforcement in 2011, the results showed that Louisiana ranked bottom failing in all quarters of implementing the Clean Air Act *(Castellón 2021)*. Now when the EPA establishes the NCAA with the industry, as it cannot eliminate every single practice conducted by the industry, it has to take into account the cost of compliance for industrial companies. Now what if the industry does not comply with the EPA assessment, the industry might take EPA to court and reverse the decision, or if the industry takes EPA court in the meantime, it could rise the cost of technology which could lead to a butterfly effect such as rise in prices because of higher cost of production, changing the economy for that particular area. On the other hand, there are some concerns on how EPA takes data whether there has been exceeding the limit of safety levels of pollution. The EPA calculates the limits by average on percentile levels. Instead of looking at the levels as a whole in different areas, the average limitations deems some areas as safe whereas it might not be safe and allows CAA violations to go unchecked. (Castellón 2021)

To understand how petrochemical and other waste producing companies were established in Louisiana, it was due to the fact of the Louisiana Department of Environmental Quality being lenient in their case of giving out permits. Instead of fighting for public health, LDEQ justified the decision arguing the benefits of social and economical gain were far greater than environmental impacts. Furthermore, companies paid LDEQ employees with overtime fees to expedite the process of construction of those facilities. LDEQ seems to favor alot of industrial gains over the public health of the community and then, governors are no exception to it. One of the governors offered a tax exemption of $130 million over a period of ten years to move the industry to Convent *(Castellón 2021)***.** Moreover, Congress enacted RCRA to manage the permitting process but as it was known the public cannot overturn the decision until the state environment agency has made a decision. To make matters worse, even if the public was involved in the process of overturning the decision, understanding and competing with the permitting process require intensive legal and technical financial resources that the poor and people of color may not have. In addition, the violations given to companies in exceeding the safety limit especially in white dominant and affluent communities were much higher than the areas of low income communities *(Castellón 2021)*

Overall, the EPA and other environmental agencies have failed to protect the people of color mainly African Americans living next to the Mississippi River, whereas to make matters worse, EPA's fight for the people has dwindled, for better understanding of how the demographics and toxins are released also exposure pathways, we will discuss it in the next section.

Demographics and Analysis of Chemical pollution:

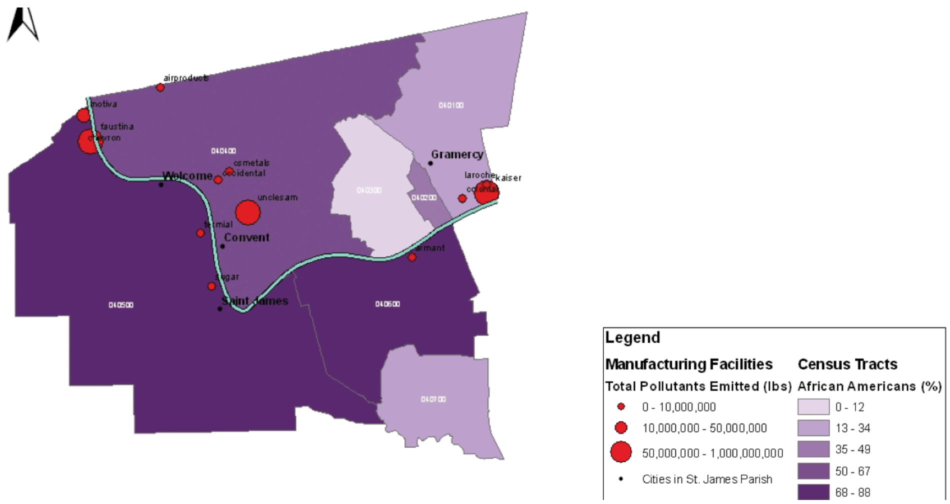


Figure 1: *Source: Nelson Mandela School ofPublic Policy and Urban Affairs Southern University and A & M College.652A. D. Blodgett*

As we can see from Figure 1, the majority of the communities that are affected by the pollution of the toxic waste companies are predominantly African American communities. (Blodgett, 2007)

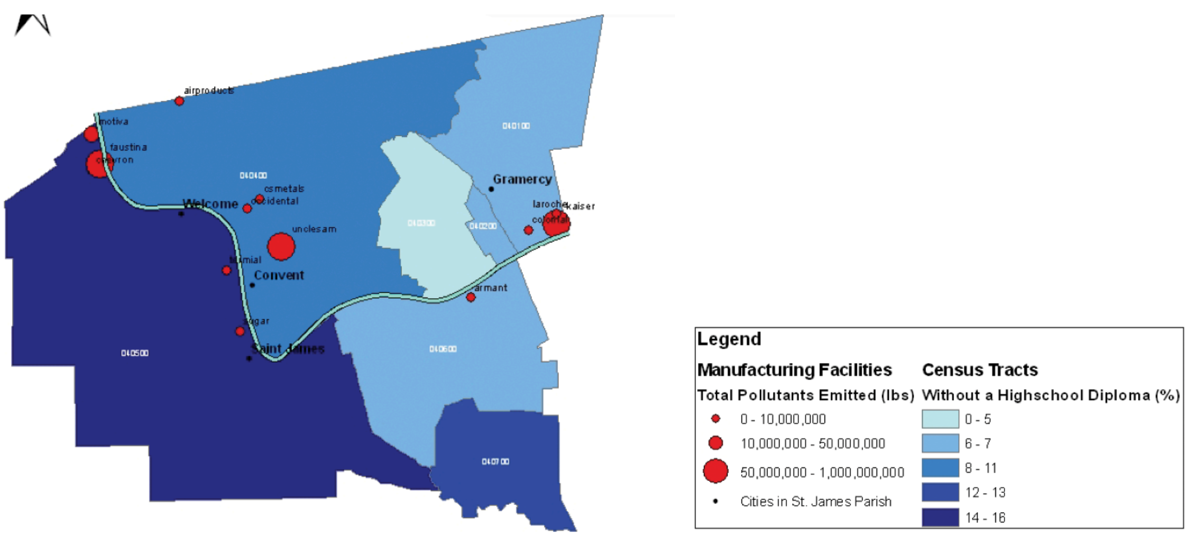
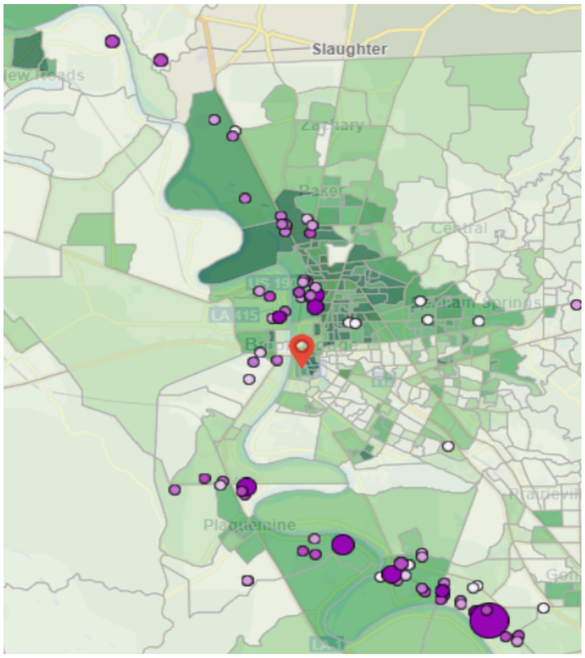


Figure 2: *Source: Nelson Mandela School ofPublic Policy and Urban Affairs Southern University and A & M College.652A. D. Blodgett*

It was found in the journal article written by Abigail D.Blodgett in "*An Analysis of Pollution and Community Advocacy in 'Cancer Alley': Setting an Example for the Environmental Justice Movement in St James Parish, Louisiana*" that majority of the manufacturing facilities in St. James Parish constructed were in the communities with the least institutional education. The study also further detailed that the land also hosted five of the 13 major facilities including the most pollutant one IMC Phosphates Faustina Plant. *(Blodgett 2007).* In the next few pages, we are using the data toolset form the TRI to understand people of color demographics and the toxins released in a recent year and over a period of 10 years to understand the accumulation of risk diseases from the toxins.



National Percentiles for % People of Color

National Percentiles	Percentages
95 - 100 percentile	≥ 97.56
90 - 95 percentile	< 97.56
80 - 90 percentile	< 91.07
70 - 80 percentile	< 72.83
60 - 70 percentile	< 54.96
50 - 60 percentile	< 40.74
Less than 50 percentile	< 29.71

Figure 3: *Source: TRI Report (Baton Rouge next to Mississippi River) of chemical releases in 2022*

In Figure 3, we can see the majority of the chemical releases are still ongoing in recent times within the regions of people of color, as this might be shocking, the practices still continue to this day. (EPA TRI, n.d.)

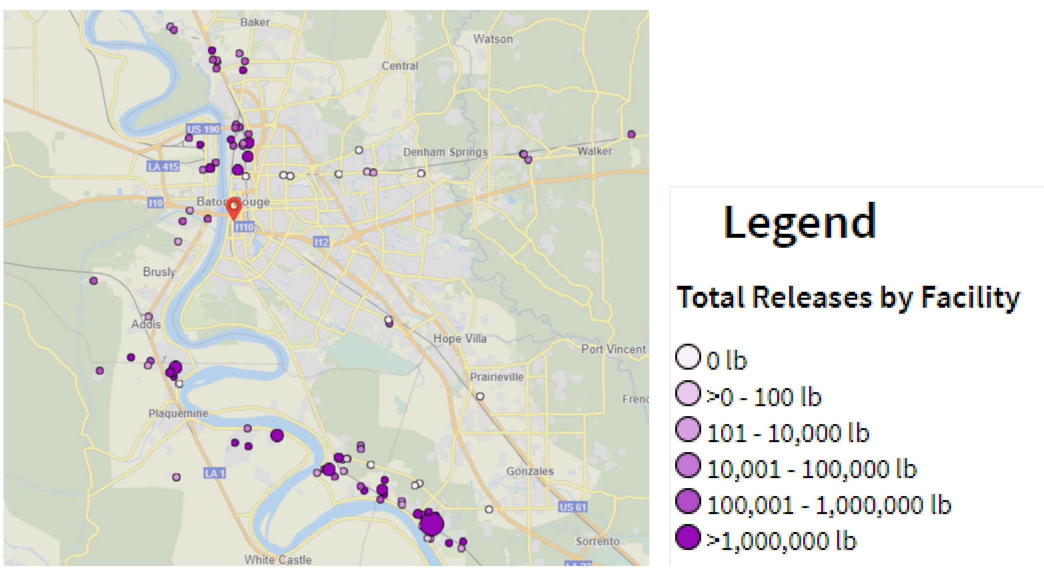


Figure 4: *Source: TRI Report (Baton Rouge next to Mississippi River) of chemical releases over a period of 10 years from 2013 - 2022*

As you can see from Figure 4, the majority of facilities chemical releases are next to the Mississippi River with an amount of more than 1,000,000 lbs across the Baton Rouge line over a period of 10 years hosting about 30,117 facilities. Furthermore, what is astonishing is that next to White Castle is a petrochemical Rubicon LLC which has high releases of more than 1,000,000 lbs, one of the major contributing chemical releases to the Mississippi River. (EPA TRI, n.d.)

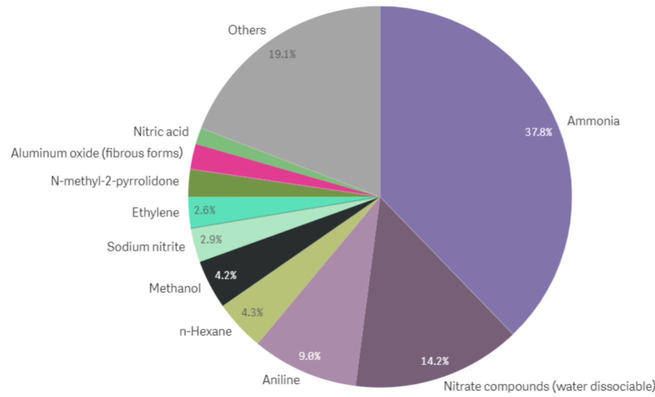


Figure 5: *Source: TRI Report (Baton Rouge next to Mississippi River) of chemical releases list*

From Figure 5, there are the top 10 chemical releases from the TRI report that state the ammonia to be the major release of 37% from the 42,577,850 lbs, whereas from the lowest it is Benzyl Chloride release which is 0.0000002% from the 42,577,850 lbs to the Mississippi River. (EPA TRI, n.d.)

Chemicals Pollution Impact on human health in Baton Rouge:

Note: *All the chemical causes are in the reference sources under either Center of Diseases, National Health and other government reference websites.*

Chemical Name	Potential Causes to Human Health	Exposure Pathway
Ammonia	*Irritation burns skin, throat, eyes, mouth, and can cause damage to lungs that lead to death.*	*From everywhere, soil, air, land mostly from indigestion*
Nitrate Compounds	*NOC are carcinogens and teratogens (causing various malformations)*	*Drinking water and dietary sources*
Aniline	*Oxidation of hemoglobin that causes headaches, tremor, heart attack and possible death*	*Inhalation, Ingestion and dermal absorption*
n-Hexane	*Damage to the nervous system*	*Air*
Methanol	*Extremely toxic, causes ataxia, heart and lung damages and death*	*Ingestions, skin absorption and inhalation*
Sodium Nitrite	*Headaches, abdominal cramps, vomiting, and death*	*Ingestion and Inhalation, drinking water*
Ethylene	*Neurological disorders, bronchitis, and vomiting*	*Inhalation and dermal contact*
Aluminium Oxide	*Lung failure and neurological damages*	*Inhalation, drinking water and road dust*
N-methyl-2-pyrrolidone	*Unconsciousness and irritation to the eyes*	*Inhalation, skin absorption and ingestion*

Nitric Acid	*Highly corrosive, bronchitis, dental erosion*	*Ingestion and inhalation*

As the Mississippi River passes through New Orleans, we will look into more data from the TRI to be further away from the horrific impacts of these chemical releases.

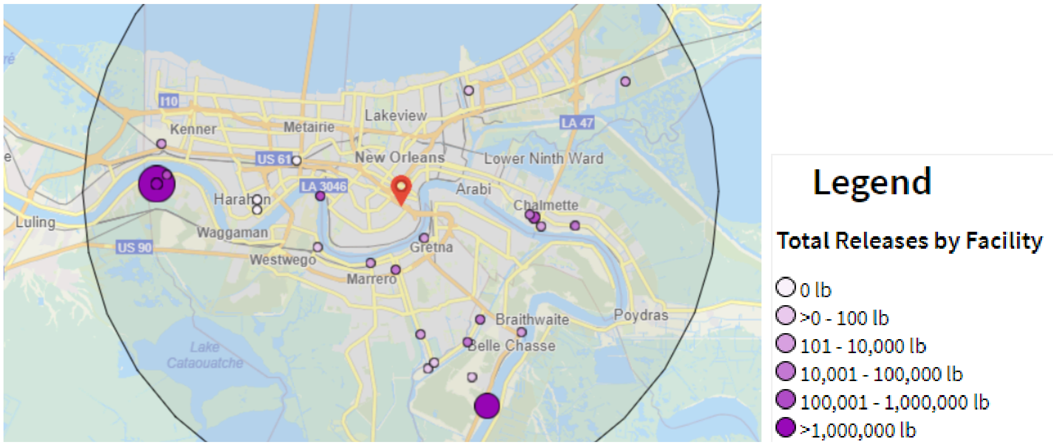


Figure 6: *Source: TRI Report (New Orleans next to Mississippi River) of chemical releases in the year 2022*

In Figure 6, the one big purple dot next to Kenner is a chemical factory called Rohem American LLC which is releasing chemical releases to the Mississippi river stream. From the TRI website, there are a total of 21,752 facilities and 522 chemicals in a 15 mile radius reported in the year 2022. (EPA TRI, n.d.)

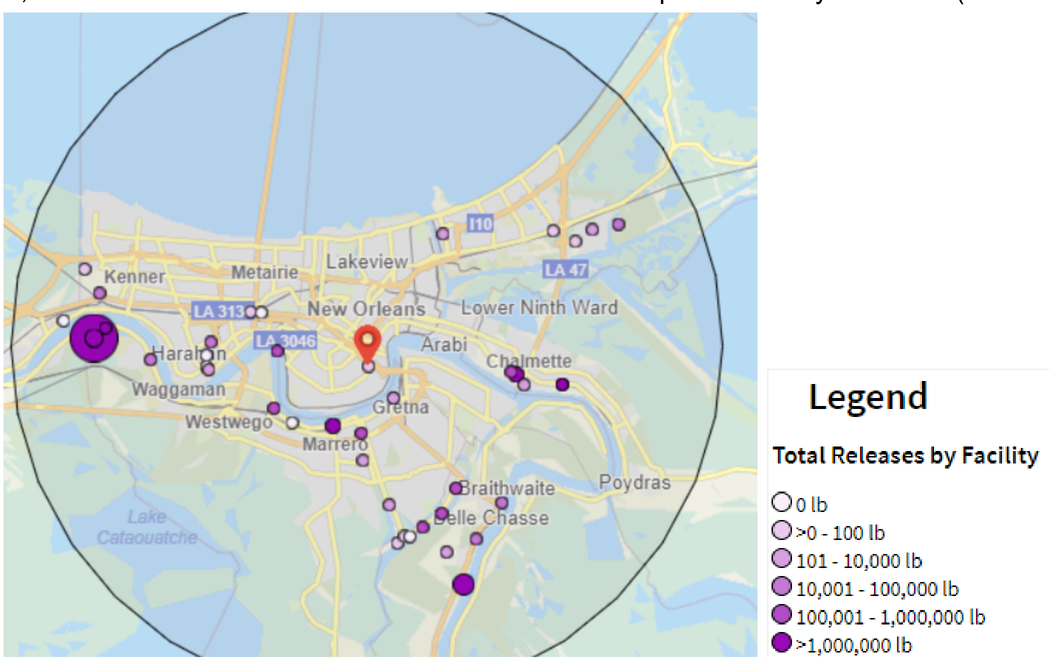


Figure 7: *Source: TRI Report (New Orleans next to Mississippi River) of chemical releases over the period of 10 years from 2013 - 2022*

In Figure 7, New Orleans reported 30,117 Facilities and 616 chemicals reported over a span of 10 years from 2013 to 2022. Now in the year of 2022, the only two biggest facilities still participating in the chemical release from the 10 year are Rohem American LLC and Oak Oil plant (the biggest purple dot right beside the Belle Chasse). (EPA TRI, n.d.)

National Percentiles for % People of Color

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95 - 100 percentile	≥ 97.56
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Figure 8: *Source: TRI Report (New Orleans next to Mississippi River) of People of color areas where the facilities are located*

The Figure 8 describes the people of color in New Orleans supplementing the situation of where the people of color communities live. That is where the majority of the chemical waste producing facilities are also stationed.

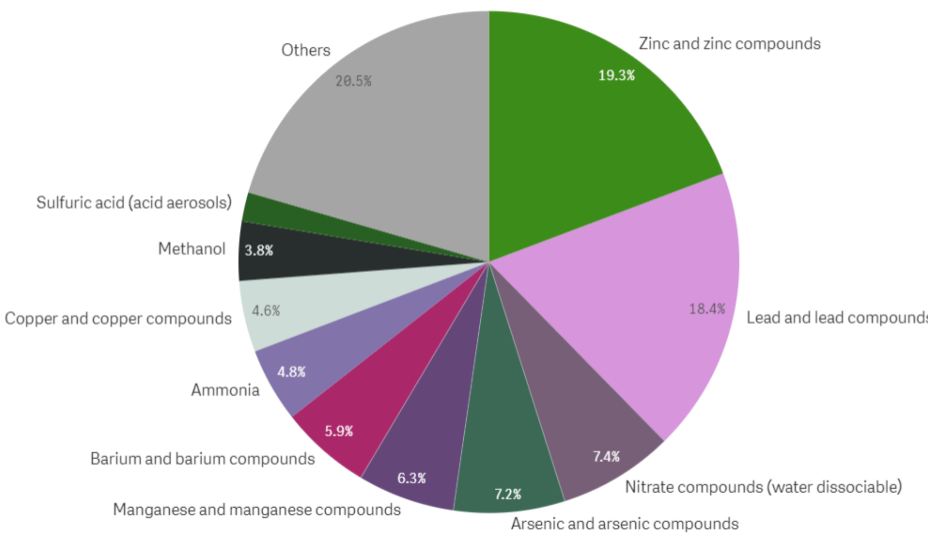

Figure 9: *Source: TRI Report (New Orleans next to Mississippi River) chemical releases over a period of 10 years*

From Figure 9, it is illustrated that the other chemicals take the lead by 20% followed by the zinc compounds by 19.3%, and finally, the zinc is followed by lead compounds which are up to 18% over a period of 10 years.

Chemicals Pollution Impact on human health in New Orleans:

Chemical Name	Potential Causes to Human Health	Exposure Pathway
Zinc	*Nausea, dizziness, headaches, and low immunity*	*Inhalation, oral and dermal*
Lead	*Damage to nervous systems and permanent intellectual disability in children*	*Inhalation and ingestion; groundwater*
Arsenic	*Heart diseases and induces cancer*	*Inhalation and ingestion; fossil fuels*
Manganese	*Damage kidneys, impact reproduction and inhibit nervous systems*	*Inhalation and ingestion ; groundwater*
Barium	*Paralysis, muscle weakness and weak gut*	*Contaminated water*
Copper	*Ingestions problems such as vomiting and pain*	*Inhalation*

Note: *All the chemical causes are in the reference sources under either Center of Diseases, National Health and other government reference websites.*

Impacts in the Cancer Alley:

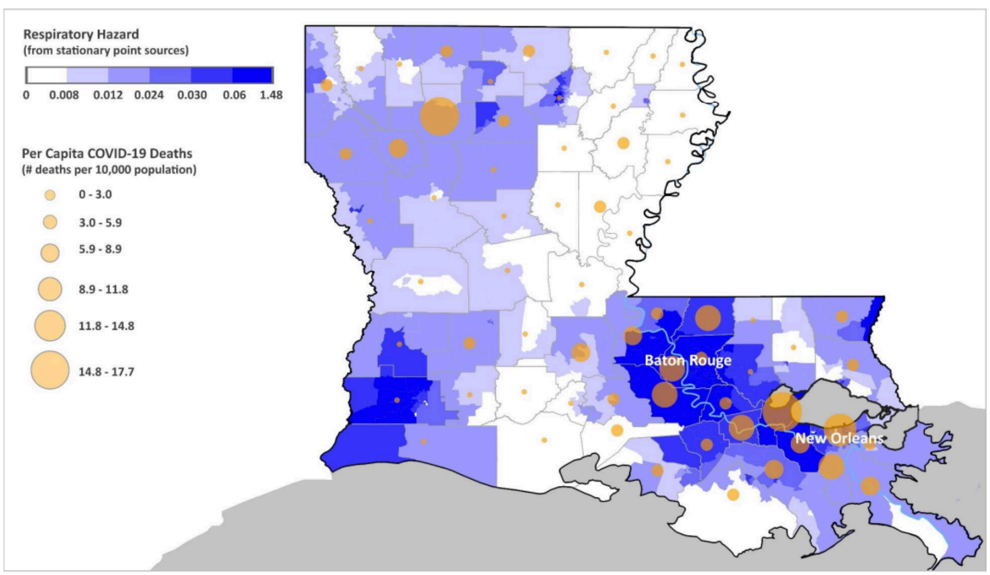

Figure 10: *Source: Respiratory Hazard by Wesley James*

A lot of the cancer alley have suffered from the respiratory hazard in the journal article from "*Air Pollution and COVID-19: A Double Whammy for African American and Impoverished Communities in Cancer Alley*" by Wesley James, summarized brilliantly that the Louisiana facing the unhygienic health from the production of the ever rising air emissions by industrial sources. (Terell and James)

DEMOGRAPHIC VARIABLE	POLLUTION ESTIMATE									
	Long-Term $PM_{2.5}$[**]		Respiratory Hazard[†]				Immunological Hazard[†]			
			All Sources		Point Source		All Sources		Point Source	
	r	P	r	P	r	P	r	P	r	P
% Black	0.25	<0.0001	0.14	< 0.0001	0.15	< 0.0001	0.14	< 0.0001	NA	0.87
% White	-0.28	<0.0001	-0.11	0.0001	-0.14	< 0.0001	-0.16	< 0.0001	NA	0.74
% 65 yrs+	-0.10	<0.001	-0.13	< 0.0001	-0.09	0.002	-0.13	< 0.0001	-0.08	0.004
% Unemployment	0.16	<0.0001	0.12	< 0.0001	0.11	0.001	0.09	0.003	NA	0.84
% Poverty	0.09	0.003	NA	0.30	NA	0.41	NA	0.60	-0.06	0.042
% Seniors in Poverty	0.12	<0.0001	NA	0.91	NA	0.94	NA	0.74	-0.08	0.009

Table 3: *Source: pollution estimate Terell and James*

The table dictated from the pollution estimate by Terell and James summarized their findings, that the majority of the demographic people who were unemployed and black have higher changes susceptible to the respiratory hazard compared to the other group. (Terell and James)

Fight for survival:

Since the rise of the facilities increased across the region of New Orleans and Baton Rouge, the need for survival increased that led to the Great Louisiana Toxins March that took place in November 1988. In the St.James parish, the group called “Rise St.James Parish” , a local community organization, is battling to stop new proposals for a chemical plant from Formosa Petrochemical. A recent article by Brian Roewe in “*Activists in Louisiana's 'Cancer Alley' hail halt to petrochemical complexes*” summarized that the environmental activist Rise St.James were successful in overturning a decision for the proposal of the new chemical plant by the Formosa petrochemical that will exceed not only state but also federal limits of releases toxins in 2020. (Roewe)

However, the same cannot be said for the EPA, according to the article “*THE EPA IS BACKING DOWN FROM ENVIRONMENTAL JUSTICE CASES NATIONWIDE*” by Delany Nolan stated that the EPA are backing down from all environmental justice in 2024, as it is truly shocking, the reason behind this effect was due to Landry suit suing the EPA program under the pretense of “*dystopian nightmare and social justice warriors fixated on rac*e”, this led to EPA dropping case for Title VI of civil rights. The EPA's decision to halt these investigations was probably influenced by concerns that Jeff Landry's lawsuit might reach the conservative-majority Supreme Court, which has shown a tendency to restrict the EPA's regulatory powers. Recent rulings by the Supreme Court have already limited the EPA's authority, suggesting that a decision on Landry's lawsuit could weaken the specific enforcement tool in question and inhibit federal agencies' capacity to uphold civil rights protections. This apprehension was increased by the Supreme Court's recent ruling against affirmative action, which indicated a critical stance on the role of race in federal policy-making. This retreat caused negative lingering effects as it cut ties off any future federal support. (Nolan)

It pains me to write this but how far a protest must go to such a level that it has to convince the governors and elected officials to make dramatic changes in a shorter period of time for the people of color. This was seen in the 1979 Love Canal residents protest, where the protestors had to go to extreme measures, lighting things up on fire, damaging the property, and making the city come to a standstill. Then, and only, then the changes were made such as relocation to a paid home by the governors. (Friese 2023)

Conclusion:

The cancer alley residents have suffered from time to time again regarding the toxins released from the petrochemical regime. At this point people will start wondering whether to move out of those red zones or continue fighting for the survival of your place. Cancer alley portrays the longest era of petrochemicals and other companies for environmental racism and abuse of power which the EPA and LDEQ are not going to stand for the safety of people but their own interest. Sadly, the end note of this case study can be summarized to one sentence when local officials quietly changed District 5 from residential occupation to residential/future industrial.

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